THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a quadrilateral, a diagonal of 30 cm has perpendiculars of 7 cm and 3 cm drawn to it from the opposite vertices. Its area is: 1

(a) 40 cm² **(b)** 150 cm² **(c)** 300 cm² **(d)** 105 cm²

2. The cost of flooring a 30 m by 15 m hall at ₹ 35 per square metre is (in ₹): 1

(a) 15750 **(b)** 485 **(c)** 1575 **(d)** 3150

3. 7 m² expressed in cm² is: 1

(a) 700 **(b)** 700000 **(c)** 70000 **(d)** 7000

4. The area of a trapezium is 108 cm². One parallel side is 11 cm and the height is 12 cm. The other parallel side (in cm) is: 1

(a) 9 **(b)** 18 **(c)** 7 **(d)** 4

5. The area of a rectangle is 532 cm² and its length is 28 cm. Its breadth (in cm) is: 1

(a) 504 **(b)** 21 **(c)** 266 **(d)** 19

6. The area of a triangle is 120 cm² and its base is 30 cm. Its height (in cm) is: 1

(a) 10 **(b)** 8 **(c)** 11 **(d)** 16

7. In a quadrilateral, a diagonal of 16 cm has perpendiculars of 9 cm and 5 cm drawn to it from the opposite vertices. Its area is: 1

(a) 112 cm² **(b)** 30 cm² **(c)** 72 cm² **(d)** 224 cm²

8. The cost of flooring a 16 m by 11 m hall at ₹ 75 per square metre is (in ₹): 1

(a) 2025 **(b)** 13200 **(c)** 251 **(d)** 4050

9. 12 m² expressed in cm² is: 1

(a) 1200000 **(b)** 12000 **(c)** 1200 **(d)** 120000

10. The area of a trapezium is 126 cm². One parallel side is 18 cm and the height is 9 cm. The other parallel side (in cm) is: 1

(a) 8 **(b)** 12 **(c)** 28 **(d)** 10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A quadrilateral has a diagonal of 22 cm and the perpendiculars from the other two vertices to it are 7 cm and 10 cm. Find its area. 2

12. The area of a rectangular field is 960 m² and its length is 64 m. Find its breadth and perimeter. 2

13. Find the area of a parallelogram whose base is 300 cm and corresponding height is 200 cm. Express it in m². 2

14. A quadrilateral has a diagonal of 40 cm and the perpendiculars from the other two vertices to it are 11 cm and 4 cm. Find its area. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangular lawn is 50 m by 16 m. Find its area, and the cost of levelling it at ₹ 9 per m². 3

16. The parallel sides of a trapezium are 31 cm and 19 cm and its area is 300 cm². Find the distance between the parallel sides. 3

17. The diagonals of a rhombus are 24 cm and 32 cm. Find its area and its side. 3

OR

A rectangular lawn is 42 m by 15 m. Find its area, and the cost of levelling it at ₹ 14 per m². 3

18. The parallel sides of a trapezium are 21 cm and 7 cm and its area is 224 cm². Find the distance between the parallel sides. 3

OR

The diagonals of a rhombus are 14 cm and 48 cm. Find its area and its side. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A trapezium has parallel sides 24 cm and 16 cm and the distance between them is 11 cm. Find its area. A parallelogram of the same height and base 16 cm (the shorter parallel side) is cut from it, leaving a triangle: find the area left over. 5

OR

A rectangular room 16 m by 15 m is to be carpeted with a carpet 6 m wide costing ₹ 60 per metre. Find the length of carpet needed and the total cost. 5

20. A trapezium has parallel sides 22 cm and 6 cm and the distance between them is 10 cm. Find its area. A parallelogram of the same height and base 6 cm (the shorter parallel side) is cut from it, leaving a triangle: find the area left over. 5

OR

A rectangular room 20 m by 9 m is to be carpeted with a carpet 5 m wide costing ₹ 150 per metre. Find the length of carpet needed and the total cost. 5

END OF QUESTION PAPER — Total: 40 Marks